Ex: NO 2

Date: 23/7/26

Study of different types of network cables.

AIM:

Study of different of network cables.

Cable types

1. unshield twisted pair cable.

Catogory 3 can transmit 10mbps data.

Catory 5 upto 100 mpbs.

catogory 5e 1 Gbps.

advantage:

(i) cheaper cost (ii) easy to install & have overall small diameter.

disadvatge:

More prone to EMI

applications:

(i) 10Base - T Ethernet. (ii) Fast Ethernet, Gigabit Ethernet.

2. Shield. twisted pair cable.

Category 6, 6a can transmit 10 Gbps.

Catogory 7 cain transmit 10Gbps.

advantge:

(i) shield (ii) Faster then UTP. (iii) Less susceptionable to noise.

disadvantge:

(i) Expansive (ii) Greater installition effort.

Appliation

(i) widely used in data centre. 10G ethernet. (55 m)

(ii) Gigabith Ethernet 10 G ethernet (100m)

3) coaxial cable. RG-6, RG-59, RG-11.

(i) can transmit 10-100 Mbps.

advantage:

(i) High bandwidth. (ii) Immune to interface. iii) versatile

(iv) Low loss · bandwidth.

disadvantages.

i) Limited distance. (ii) cost (iii) Size is bulky.

Application.

speed of signal is 500 m television network High speed. internet connections.

4) fibre optics cable.

single mode & multimode. can transmit over. 100 Gbps.

advantages:

(i) High speed. (ii) High bandwidth. (iii) High security.

(iv) Long distance.

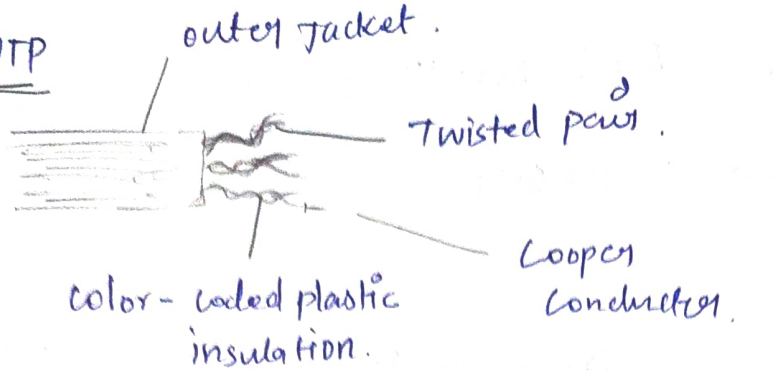
disadvantages.

(i) Expensive. (ii) Requires skilled. installers.

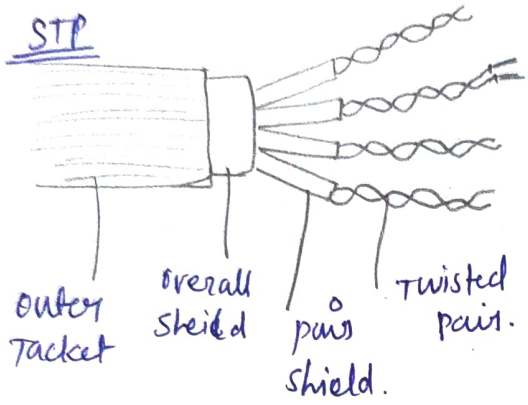
Application:

Maximum distance of fibre optics cable is around 100 meters.

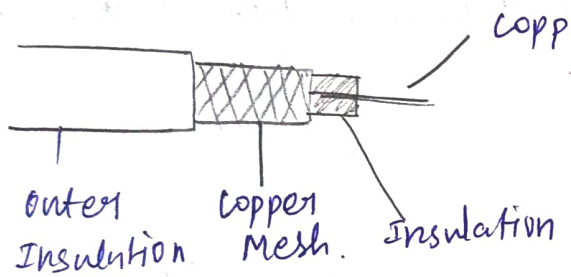
1. UTP



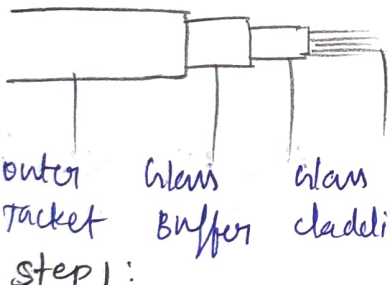
2. STP



3.



4.



Step 1:

1. To start construction of the device, begin by threading shields onto cable.

Step 2:

2. Next strip approximetly 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3:

3. After you will need to untangle wires. there should be four twisted pairs referig back

to the sheet, arrage them from top to bottom. one end should be in arrangement A and the other B.

Step4:

Trim the aligned wires flat then insert them into the RJ45 plug with the clip facing away and gold pins facing you.

Step5:

push the cable into the plug so the notch sits over the plug firmly using a crimping tool to secure the wires and cable in place.

Step6:

Crimp the RJ45 plug firmly using a crimply tool to secure the wires and cable in place.

Step7:

Repeat the process for the opposite end of the opposite Cable using the desired pin.

Step8:

Test the completed cable using a LAN cable tester to ensure proper connectivity across all pins.

Result:

Network cable types were Studied and Stright through and cross - wired cable were successfully constructed and tested using a crimping tool and LAN tester.